

Quiz-3 (DS412)

Attempt all questions. MM:40, Time: 50 Minutes

1. Construct four matrices $\mathbf{A}_r, r \in \{1, 2, 3, 4\}$ (rank r & size $m \times n$) listed the possibilities for $\mathbf{A}_r \mathbf{x} = \mathbf{b}$ below:

(a) $r = m = n$ and $\mathbf{A}_1 \mathbf{x} = \mathbf{b}$ has 1 solution for every $\mathbf{b}$. 2.5

(b) $r = m < n$ and $\mathbf{A}_2 \mathbf{x} = \mathbf{b}$ has 1 solution for every $\mathbf{b}$. 2.5

(c) $r = n < m$ and $\mathbf{A}_3 \mathbf{x} = \mathbf{b}$ has no solution. 2.5

(d) $r < m, r < n$ and $\mathbf{A}_4 \mathbf{x} = \mathbf{b}$ has 0 or ∞ solutions. 2.5

2. Answer the below parts when the matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ follow $\mathbf{K} = \mathbf{ABC}$, given below.

$$\mathbf{K} = \underbrace{\begin{bmatrix} 1/\sqrt{1} & 1/\sqrt{4} & 1/\sqrt{4} & 1/\sqrt{4} \\ 1/\sqrt{1} & 1/\sqrt{1} & -1/\sqrt{4} & -1/\sqrt{1} \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}}_{\mathbf{B}} \underbrace{\begin{bmatrix} 1/\sqrt{4} & 1/\sqrt{4} & 1/\sqrt{2} & 0 \\ 1/\sqrt{1} & 1/\sqrt{1} & 1/\sqrt{2} & 0 \\ 1/\sqrt{1} & 1/\sqrt{1} & 0 & 1/\sqrt{2} \\ 1/\sqrt{1} & 1/\sqrt{1} & 0 & 1/\sqrt{2} \end{bmatrix}}_{\mathbf{C}}$$

$\mathbf{K} = \mathbf{M} \mathbf{\Lambda} \mathbf{M}^T$

(a) Observe (don't multiply/ solve) and tell the determinant of $\mathbf{K}$. Explain your answer. 2

(b) Observe (don't multiply/ solve) and tell the eigenvalues of $\mathbf{K}$. Explain your answer. 2

(c) Observe (don't multiply/ solve) and tell the eigenvectors of $\mathbf{K}$. Explain your answer. 2

(d) Let the SVD of $\mathbf{K} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$, then obtain $\mathbf{U}$, $\mathbf{\Sigma}$, $\mathbf{V}$. 5

(e) Answer the parts 2a-2c for $\mathbf{K}_1$ when $\mathbf{K}_1 = \mathbf{ACB}$. 5

(f) Can $\mathbf{K}_1$ in 2c be factored as $\mathbf{K}_1 = \mathbf{DD}^T$? State $\mathbf{D}$ or a reason if it can't be factored. 3

(g) State the best rank(2) approximation for $\mathbf{K}$. 5

(h) Obtain $\mathbf{K}^{-1}$. Show all the steps. 6